



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201

ENGINEERING MATHEMATIC

UNIT IV

VECTOR INTEGRAL CALCULUS

1. STOKES' THEOREM

Stokes' Theorem can be regarded as a higher-dimensional version of Green's Theorem.

Green's Theorem relates a double integral over a plane region D to a line integral around its plane boundary curve.

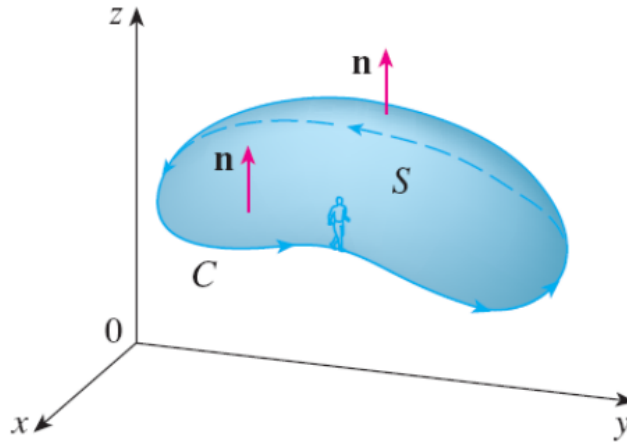
Stokes' Theorem relates a surface integral over a surface S to a line integral around the boundary curve of S (a space curve).

1.1 INTRODUCTION

Oriented surface with unit normal vector n .

The orientation of S induces the positive orientation of the boundary curve C .

If you walk in the positive direction around C with your head pointing in the direction of n , the surface will always be on your left.



The theorem is named after the Irish mathematical physicist Sir George Stokes (1819–1903).

What we call Stokes' Theorem was actually discovered by the Scottish physicist Sir William Thomson (1824–1907, known as Lord Kelvin).

Stokes learned of it in a letter from Thomson in 1850.

Let S be an oriented piecewise-smooth surface bounded by a simple, closed, piecewise-smooth boundary curve C with positive orientation.

F be a vector field whose components have continuous partial derivatives on an open region in R_3 that contains S . Then,

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} dS$$

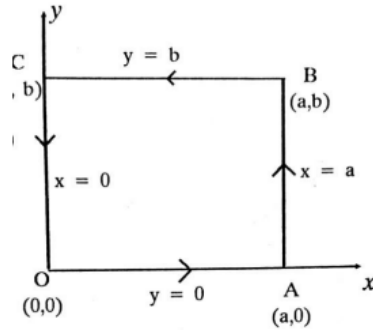
$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \nabla \times \vec{F} \cdot \hat{n} ds$$

The line integral around the boundary curve of S of the tangential component of F is equal to the surface integral of the normal component of the curl of F .

1.2 Problems based on Stoke's theorem:

1. Verify stokes theorem for a vector field defined by $\vec{F} = (x^2 - y^2)\vec{i} + 2xy\vec{j}$ in a rectangular region in the xoy plane bounded by the lines $x = 0, x = a, y = 0, y = b$.

Solution:



By Stokes theorem, $\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} dS$

To evaluate: $\iint_S \text{Curl } \vec{F} \cdot \hat{n} dS$

Given $\vec{F} = (x^2 - y^2)\vec{i} + 2xy\vec{j}$

$\text{Curl } \vec{F} = \nabla \times \vec{F}$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 - y^2 & 2xy & 0 \end{vmatrix} = 4y\vec{k}$$

Since the surface is a rectangle in the xy plane, $\hat{n} = \vec{k}$, $dS = dxdy$

$\text{Curl } \vec{F} \cdot \hat{n} = 4y\vec{k} \cdot \vec{k} = 4y$, Order of integration is $dxdy$

x varies from $x = 0$ to $x = a$

y varies from $y = 0$ to $y = b$

$$\begin{aligned} \iint_S \text{Curl } \vec{F} \cdot \hat{n} dS &= \int_0^b \int_0^a 4y dxdy \\ &= \left[\frac{4ay^2}{2} \right]_0^b = 2ab^2 \end{aligned} \quad \text{-----(1)}$$

To evaluate: $\int_C \vec{F} \cdot d\vec{r}$

Here the line integral over the simple closed curve C bounding the surface $OABCO$ consisting of the edges OA , AB , and CO .

Curve	Equation	Limit
OA	$y = 0$	$x = 0$ to $x = a$
AB	$x = a$	$y = 0$ to $y = b$
BC	$y = b$	$x = a$ to $x = 0$
CO	$x = 0$	$y = b$ to $y = 0$

Therefore, $\int_C \vec{F} \cdot d\vec{r} = \int_{OA} + \int_{AB} + \int_{BC} + \int_{CO}$

$$\vec{F} \cdot d\vec{r} = (x^2 - y^2)dx + 2xydy \quad \text{-----}(2)$$

On OA: $y = 0$, $dy = 0$, x varies from 0 to a

$$(2) \Rightarrow \int_{OA} \vec{F} \cdot d\vec{r} = \int_0^a x^2 dx = \left[\frac{x^3}{3} \right]_0^a = \frac{a^3}{3}$$

On AB: $x = a$, $dx = 0$, y varies from 0 to b

$$(2) \Rightarrow \int_{AB} \vec{F} \cdot d\vec{r} = \int_0^b 2ay dy = \left[\frac{2ay^2}{2} \right]_0^b = ab^2$$

On BC: $y = b$, $dy = 0$, x varies from a to 0

$$(2) \Rightarrow \int_{BC} \vec{F} \cdot d\vec{r} = \int_a^0 x^2 - b^2 dx = \left[\frac{x^3}{3} - b^2 x \right]_a^0 = -\frac{a^3}{3} + ab^2$$

On CO: $x = 0$, $dx = 0$, y varies from b to 0

$$(2) \Rightarrow \int_{CO} \vec{F} \cdot d\vec{r} = 0$$

$$\therefore \int_C \vec{F} \cdot d\vec{r} = \frac{a^3}{3} + ab^2 - \frac{a^3}{3} + ab^2 = 2ab^2 \quad \text{-----}(3)$$

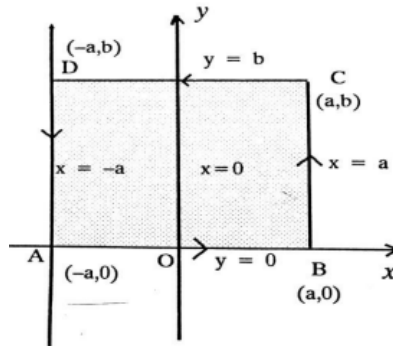
$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} dS$$

From (3) and (1)

Hence Stokes theorem is verified.

2. Verify Stoke's theorem for $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$ taken around the rectangle bounded by the lines $x = \pm a$, $y = 0$, $y = b$.

Solution:



$$\text{By Stokes theorem, } \int_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} dS$$

$$\iint_S \text{Curl } \vec{F} \cdot \hat{n} dS$$

To evaluate:

$$\text{Given } \vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$$

$$\text{Curl } \vec{F} = \nabla \times \vec{F}$$

$$= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix} = -4y\vec{k}$$

Since the surface is a rectangle in the xy plane, $\hat{n} = \vec{k}$, $dS = dxdy$

$\text{Curl } \vec{F} \cdot \hat{n} = -4y\vec{k} \cdot \vec{k} = -4y$, Order of integration is $dxdy$

x varies from $x = -a$ to $x = a$

y varies from $y = 0$ to $y = b$.

$$\begin{aligned} \iint_S \text{Curl } \vec{F} \cdot \hat{n} dS &= \int_0^b \int_{-a}^a -4y dxdy \\ &= -4 \int_0^b [xy]_{-a}^a dy \\ &= \left[\frac{-8ay^2}{2} \right]_0^b = -4ab^2 \end{aligned} \quad \text{-----(1)}$$

$$\int_C \vec{F} \cdot d\vec{r}$$

To evaluate:

Here the line integral over the simple closed curve C bounding the surface $ABCD$ consisting of the edges AB , BC , CD and DA .

$$\text{Therefore, } \int_C \vec{F} \cdot d\vec{r} = \int_{AB} + \int_{BC} + \int_{CD} + \int_{DA}$$

$$\vec{F} \cdot d\vec{r} = (x^2 + y^2)dx - 2xydy \quad \text{-----(2)}$$

Along AB : $y = 0$, $dy = 0$, x varies from $-a$ to a

$$\Rightarrow \int_{AB} \vec{F} \cdot d\vec{r} = \int_{-a}^a x^2 dx = \left[\frac{x^3}{3} \right]_{-a}^a = \frac{2a^3}{3} \quad (2)$$

Along BC , $x = a$, $dx = 0$, y varies from 0 to b

$$\Rightarrow \int_{BC} \vec{F} \cdot d\vec{r} = \int_0^b -2ay dy = \left[\frac{-2ay^2}{2} \right]_0^b = -ab^2 \quad (2)$$

Along CD : $y = b$, $dy = 0$, x varies from a to $-a$

$$(2) \Rightarrow \int_{CD} \vec{F} \cdot d\vec{r} = \int_{-a}^a (x^2 + b^2) dx = \left[\frac{x^3}{3} + b^2 x \right]_{-a}^a = \frac{-2a^3}{3} - 2ab^2$$

Along DC: $x = -a$, $dx = 0$, y varies from b to 0

$$(2) \Rightarrow \int_{DA} \vec{F} \cdot d\vec{r} = \int_b^0 2ay dy = \left[\frac{2ay^2}{2} \right]_b^0 = -ab^2$$

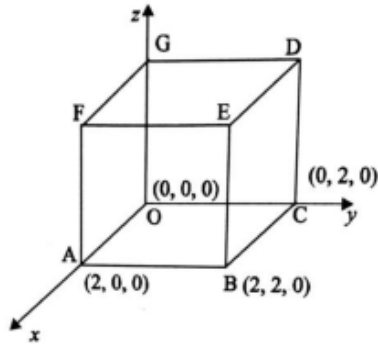
$$\therefore \int_C \vec{F} \cdot d\vec{r} = \frac{2a^3}{3} - ab^2 - \frac{2a^3}{3} - 2ab^2 - ab^2 = -4ab^2 \quad \text{-----}(3)$$

$$\text{From (3) and (1)} \quad \int_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} dS$$

Hence Stokes theorem is verified.

3. Verify Stoke's theorem in a plane for $\vec{F} = (y - z + 2)\vec{i} - (yz + 4)\vec{j} - xz\vec{k}$, where S is the open surface of the cube formed by the planes $x = 0$, $x = 2$, $y = 0$, $y = 2$, $z = 0$, $z = 2$ above the xy -plane.

Solution:



$$\text{Stoke's theorem is } \int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} dS$$

$$\text{Given } \vec{F} = (y - z + 2)\vec{i} - (yz + 4)\vec{j} - xz\vec{k}$$

$$\vec{F} \cdot d\vec{r} = (y - z + 2)dx - (yz + 4)dy - xz dz$$

$$\text{L.H.S} = \int_C \vec{F} \cdot d\vec{r} = \int_{OA} + \int_{AB} + \int_{BC} + \int_{CO}$$

In xy plane $z = 0 \Rightarrow dz = 0$

$$\vec{F} \cdot d\vec{r} = (y + 2x)dx + 4dy$$

On OA : $y = 0 \Rightarrow dy = 0$, x varies from 0 to 2.

$$\begin{aligned}\Rightarrow \int_{OA} \vec{F} \cdot d\vec{r} &= \int_0^2 2 dx \\ &= 2[x]_0^2 = 4\end{aligned}$$

On AB : $x = 2 \Rightarrow dx = 0$, y varies from 0 to 2.

$$\begin{aligned}\Rightarrow \int_{AB} \vec{F} \cdot d\vec{r} &= \int_0^2 4 dy \\ &= 4[y]_0^2 = 8\end{aligned}$$

On BC : $y = 2 \Rightarrow dy = 0$, x varies from 2 to 0.

$$\begin{aligned}\Rightarrow \int_{BC} \vec{F} \cdot d\vec{r} &= \int_2^0 4 dx \\ &= 4[x]_2^0 = -8\end{aligned}$$

On CO : $x = 0 \Rightarrow dx = 0$, y varies from 2 to 0.

$$\begin{aligned}\Rightarrow \int_{CO} \vec{F} \cdot d\vec{r} &= \int_2^0 4 dy \\ &= 4[y]_2^0 = -8\end{aligned}$$

$$\therefore \int_c \vec{F} \cdot d\vec{r} = 4 + 8 - 8 - 8 = -4 \quad \dots (1)$$

$$\text{R.H.S} = \iiint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

$$\begin{aligned}\text{curl } \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y - z + 2 & yz + 4 & -xz \end{vmatrix} \\ &= \vec{i}(0 - y) - \vec{j}(-z + 1) + \vec{k}(0 - 1) \\ &= y\vec{i} - (z - 1)\vec{j} - \vec{k}\end{aligned}$$

Given S is an open surface consisting of the 5 faces of the cube except, $xy -$ plane.

$$\iint \text{curl } \vec{F} \cdot \hat{n} \, ds = \iint_{S_1} + \iint_{S_2} + \dots + \iint_{S_5}$$

$$\text{curl } \vec{F} = y\vec{i} - (z - 1)\vec{j} - \vec{k}$$

Faces	Plane	ds	$\hat{n}$	$\text{curl } \vec{F} \cdot \hat{n}$	
Top (S_1)	xy	$dxdy$	$\vec{k}$	-1	$\int_0^2 \int_0^2 -1 \, dxdy$
Left (S_2)	xz	$dxdz$	$-\vec{j}$	$-(z-1)$	$\int_0^2 \int_0^2 (-z+1) \, dxdz$
Right (S_3)	xz	$dxdz$	$\vec{j}$	$(z-1)$	$\int_0^2 \int_0^2 (z-1) \, dxdz$
Back (S_4)	yz	$dydz$	$-\vec{i}$	y	$\int_0^2 \int_0^2 y \, dydz$
Front (S_5)	yz	$dydz$	$\vec{i}$	$-y$	$\int_0^2 \int_0^2 -y \, dydz$

$$\begin{aligned}
\text{On } S_1: \int_0^2 \int_0^2 (-1) \, dxdy \\
&= - \int_0^2 [x]_0^2 \, dy \\
&= -2 \int_0^2 dy \\
&= -2[y]_0^2 = -4
\end{aligned}$$

$$\begin{aligned}
\text{On } S_2: \int_0^2 \int_0^2 (-z+1) \, dxdz \\
&= \int_0^2 (-z+1)[x]_0^2 \, dz \\
&= 2 \int_0^2 (-z+1) \, dz \\
&= 2 \left[-\frac{z^2}{2} + z \right]_0^2 = 2(0) = 0
\end{aligned}$$

$$\begin{aligned}
\text{On } S_3: \int_0^2 \int_0^2 (z-1) \, dxdz \\
&= \int_0^2 (z-1)[x]_0^2 \, dz \\
&= 2 \int_0^2 (z-1) \, dz \\
&= 2 \left[\frac{z^2}{2} - z \right]_0^2 = 2(0) = 0
\end{aligned}$$

$$\begin{aligned}
 \text{On } S_4: \int_0^2 \int_0^2 y \, dy \, dz \\
 &= \int_0^2 \left[\frac{y^2}{2} \right]_0^2 dy \\
 &= 2 \int_0^2 dz \\
 &= 2[z]_0^2 = 4
 \end{aligned}$$

$$\begin{aligned}
 \text{On } S_5: \int_0^2 \int_0^2 -y \, dy \, dz \\
 &= - \int_0^2 \left[\frac{y^2}{2} \right]_0^2 dy \\
 &= -2 \int_0^2 dz \\
 &= -2[z]_0^2 = -4
 \end{aligned}$$

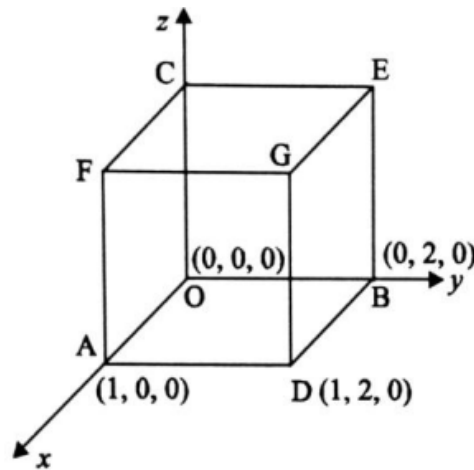
$$\therefore \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds = -4 + 0 + 0 + 4 - 4 = -4 \quad \dots (2)$$

$$\text{From (1) and (2) } \int_c \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

Hence Stoke's theorem is verified.

4. Verify Stoke's theorem in a plane for $\vec{F} = xy\vec{i} - 2yz\vec{j} - zx\vec{k}$, where S is the open surface of the rectangular parallelepiped formed by the planes $x = 0, x = 1, y = 0, y = 2, z = 0, z = 3$ above the xy -plane.

Solution:



5.

$$\text{Stoke's theorem is } \int_c \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

$$\text{Given } \vec{F} = xy\vec{i} - 2yz\vec{j} - zx\vec{k}$$

$$\vec{F} \cdot d\vec{r} = xydx - 2yzdy - zx \, dz$$

$$\text{L.H.S} = \int_c \vec{F} \cdot d\vec{r} = \int_{OA} + \int_{AB} + \int_{BC} + \int_{CO}$$

$$\text{In } xy \text{ plane } z = 0 \Rightarrow dz = 0$$

$$\vec{F} \cdot d\vec{r} = xydx$$

$$\text{On } OA: y = 0 \Rightarrow dy = 0, x \text{ varies from } 0 \text{ to } 1.$$

$$\begin{aligned} \Rightarrow \int_{OA} \vec{F} \cdot d\vec{r} &= \int_0^1 0 \, dx \\ &= 0 \end{aligned}$$

$$\text{On } AB: x = 1 \Rightarrow dx = 0, y \text{ varies from } 0 \text{ to } 2.$$

$$\begin{aligned} \Rightarrow \int_{AB} \vec{F} \cdot d\vec{r} &= \int_0^2 0 \, dy \\ &= 0 \end{aligned}$$

$$\text{On } BC: y = 2 \Rightarrow dy = 0, x \text{ varies from } 1 \text{ to } 0.$$

$$\begin{aligned} \Rightarrow \int_{BC} \vec{F} \cdot d\vec{r} &= \int_1^0 2x \, dx \\ &= 2 \left[\frac{x^2}{2} \right]_1^0 = -1 \end{aligned}$$

$$\text{On } CO: x = 0 \Rightarrow dx = 0, y \text{ varies from } 2 \text{ to } 0.$$

$$\begin{aligned} \Rightarrow \int_{CO} \vec{F} \cdot d\vec{r} &= \int_2^0 0 \, dy \\ &= 0 \end{aligned}$$

$$\therefore \int_c \vec{F} \cdot d\vec{r} = 0 + 0 - 1 + 0 = -1 \quad \dots (1)$$

$$\text{R.H.S} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

$$\text{curl } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & -2yz & -xz \end{vmatrix}$$

$$= \vec{i}(0 + 2y) - \vec{j}(-z - 0) + \vec{k}(0 - x)$$

$$= 2y\vec{i} + z\vec{j} - x\vec{k}$$

Given S is an open surface consisting of the 5 faces of the cube except, $xy -$ plane.

$$\iint \text{curl } \vec{F} \cdot \hat{n} \, ds = \iint_{S_1} + \iint_{S_2} + \dots + \iint_{S_5}$$

$$\text{curl } \vec{F} = 2y\vec{i} + z\vec{j} - x\vec{k}$$

Faces	Plane	ds	$\hat{n}$	$\text{curl } \vec{F} \cdot \hat{n}$	
Top (S_1)	xy	$dxdy$	$\vec{k}$	$-x$	$\int_0^2 \int_0^1 -x \, dxdy$
Left (S_2)	xz	$dxdz$	$-\vec{j}$	$-z$	$\int_0^3 \int_0^1 -z \, dxdz$
Right (S_3)	xz	$dxdz$	$\vec{j}$	z	$\int_0^3 \int_0^1 z \, dxdz$
Back (S_4)	yz	$dydz$	$-\vec{i}$	$-2y$	$\int_0^3 \int_0^2 -2y \, dydz$
Front (S_5)	yz	$dydz$	$\vec{i}$	$2y$	$\int_0^3 \int_0^2 2y \, dydz$

On S_1 : $\int_0^2 \int_0^1 (-1) \, dxdy$

$$= -\int_0^2 \left[\frac{x^2}{2} \right]_0^1 dy$$

$$= -\frac{1}{2} \int_0^2 dy$$

$$= -\frac{1}{2} [y]_0^2 = -1$$

$$\begin{aligned}
\text{On } S_2: \int_0^3 \int_0^1 -z \, dx \, dz &= - \int_0^3 [zx]_0^1 \, dz \\
&= - \int_0^3 z \, dz \\
&= - \left[\frac{z^2}{2} \right]_0^3 = - \frac{9}{2}
\end{aligned}$$

$$\begin{aligned}
\text{On } S_3: \int_0^3 \int_0^1 z \, dx \, dz &= \int_0^3 [zx]_0^1 \, dz \\
&= \int_0^3 z \, dz \\
&= \left[\frac{z^2}{2} \right]_0^3 = \frac{9}{2}
\end{aligned}$$

$$\begin{aligned}
\text{On } S_4: \int_0^3 \int_0^2 -2y \, dy \, dz &= -2 \int_0^3 \left[\frac{y^2}{2} \right]_0^2 \, dz \\
&= -4 \int_0^3 dz \\
&= -4[z]_0^3 = -12
\end{aligned}$$

$$\begin{aligned}
\text{On } S_5: \int_0^3 \int_0^2 2y \, dy \, dz &= 2 \int_0^3 \left[\frac{y^2}{2} \right]_0^2 \, dz \\
&= 4 \int_0^3 dz \\
&= 4[z]_0^3 = 12
\end{aligned}$$

$$\therefore \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds = -1 - \frac{9}{2} + \frac{9}{2} - 12 + 12 = -1 \quad \dots (2)$$

$$\text{From (1) and (2) } \int_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

Hence Stoke's theorem is verified.

5. Verify Stoke's theorem for $\vec{F} = y^2 z \vec{i} + z^2 x \vec{j} + x^2 y \vec{k}$, where S is the open surface of the cube formed by the planes $x = \pm a$, $y = \pm a$, and $z = \pm a$ in which the plane $z = -a$ is a cut.

Solution:

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} \, dS$$

By Stokes theorem,

$$\text{Given } \vec{F} = y^2 z \vec{i} + z^2 x \vec{j} + x^2 y \vec{k}$$

$$\vec{F} \cdot d\vec{r} = y^2 z dx + z^2 x dy + x^2 y dz$$

This square ABCD lies in the plane $z = -a \Rightarrow dz = 0$

$$\therefore \vec{F} \cdot d\vec{r} = -ay^2 dx + a^2 x dy$$

$$\text{L.H.S} = \int_c \vec{F} \cdot d\vec{r} = \int_{AB} + \int_{BC} + \int_{CD} + \int_{DA}$$

On AB: $y = -a \Rightarrow dy = 0$, x varies from $-a$ to a .

$$\begin{aligned} \Rightarrow \int_{AB} \vec{F} \cdot d\vec{r} &= \int_{-a}^a -a^3 dx \\ &= -a^3 [x]_{-a}^a \\ &= -a^3(2a) = -2a^4 \end{aligned}$$

On BC: $x = a \Rightarrow dx = 0$, y varies from $-a$ to a .

$$\begin{aligned} \Rightarrow \int_{BC} \vec{F} \cdot d\vec{r} &= \int_{-a}^a a^3 dy \\ &= a^3 [y]_{-a}^a \\ &= a^3(2a) = 2a^4 \end{aligned}$$

On CD: $y = a \Rightarrow dy = 0$, x varies from a to $-a$.

$$\begin{aligned} \Rightarrow \int_{CD} \vec{F} \cdot d\vec{r} &= \int_a^{-a} -a^3 dx \\ &= -a^3 [x]_a^{-a} \\ &= -a^3(-2a) = 2a^4 \end{aligned}$$

On DA: $x = -a \Rightarrow dx = 0$, y varies from a to $-a$.

$$\begin{aligned} \Rightarrow \int_{DA} \vec{F} \cdot d\vec{r} &= \int_a^{-a} -a^3 dy \\ &= -a^3 [y]_a^{-a} \\ &= -a^3(-2a) = 2a^4 \end{aligned}$$

$$\therefore \int_c \vec{F} \cdot d\vec{r} = -2a^4 + 2a^4 + 2a^4 + 2a^4 = 4a^4 \quad \dots (1)$$

$$\text{R.H.S} = \iint_S \text{curl } \vec{F} \cdot \hat{n} \, ds$$

$$\begin{aligned} \text{curl } \vec{F} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2z & z^2x & x^2y \end{vmatrix} \\ &= \vec{i}(x^2 - 2xz) - \vec{j}(y^2 - 2xy) + \vec{k}(z^2 - 2yz) \end{aligned}$$

Given S is an open surface consisting of the 5 faces of the cube except, $z = -a$.

$$\iint \text{curl } \vec{F} \cdot \hat{n} \, ds = \iint_{S_1} + \iint_{S_2} + \dots + \iint_{S_5}$$

$$\text{curl } \vec{F} = 2y\vec{i} + z\vec{j} - x\vec{k}$$

Faces	Plane	ds	$\hat{n}$	Eqn	$\text{curl } \vec{F} \cdot \hat{n}$	$\nabla \times \vec{F} \cdot \hat{n}$
Top (S_1)	xy	$dxdy$	$\vec{k}$	$z = a$	$z^2 - 2yz$	$a^2 - 2ay$
Left (S_2)	xz	$dxdz$	$-\vec{j}$	$y = -a$	$y^2 - 2xy$	$a^2 + 2ax$
Right (S_3)	xz	$dxdz$	$\vec{j}$	$y = a$	$-(y^2 - 2xy)$	$-(a^2 - 2ax)$
Back (S_4)	yz	$dydz$	$-\vec{i}$	$x = -a$	$-(x^2 - 2xz)$	$-(a^2 + 2az)$
Front (S_5)	yz	$dydz$	$\vec{i}$	$x = a$	$x^2 - 2xz$	$a^2 - 2az$

$$\text{On } S_1: \int_{-a}^a \int_{-a}^a (a^2 - 2ay) \, dxdy$$

$$= \int_{-a}^a [(a^2x - 2ayx)]_{-a}^a \, dy$$

$$= \int_{-a}^a (a^3 - 2a^2y) - (-a^3 + 2a^2y) \, dy$$

$$= \int_{-a}^a 2a^3 - 4a^2y \, dy$$

$$= \left[2a^3y - 4a^2 \frac{y^2}{2} \right]_{-a}^a$$

$$= (2a^4 - 2a^4) - (-2a^4 - 2a^4)$$

$$= 2a^4 - 2a^4 + 2a^4 + 2a^4$$

$$= 4a^4$$

----- (2)

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \text{Curl } \vec{F} \cdot \hat{n} \, dS$$

From (2) and (1)

Hence Stokes theorem is verified.

Practice Problems:

1. Verify Stoke's theorem for $F = (y - z + 2)\vec{i} + (yz + 4)\vec{j} - xz\vec{k}$, where S is the open surface of the cube $x = 0, y = 0, z = 0, x = 1, y = 2, z = 3$ above the xy plane.

Ans: Common value = -2

2. Verify Stoke's theorem for $F = (x^2 - y^2)\vec{i} + 2xy\vec{j} + xyz\vec{k}$, over the surface of the box bounded by the planes $x = 0, y = 0, x = a, y = b, z = c$ above the xy plane.

Ans: Common value $2ab^2$

3. Verify Stoke's theorem for the function $F = x^2\vec{i} + xy\vec{j}$, integrated round the square in the $z = 0$ plane whose sides are along the lines $x = 0, y = 0, x = a, y = a$.

Ans: $\frac{a^3}{2}$

Discussion Questions:

Stokes theorem

Try to answer the following questions

1. What is the real-life application of Stokes theorem?
2. What surface is used for Stokes theorem?
3. Is Stokes theorem used for flux?
4. What is the simplest surface of Stokes theorem?